1 STRATA Benchmark Construction

from annotated clinical timelines to temporal–statistical claims with gold at every stage

1.1 Annotated clinical timelines (MedTempo)

3,166 VAERS adverse-event reports (Janssen, Pfizer, Moderna), expert-annotated as a leveled order of events

Example timeline (report MT-0482913)

Fever → Headache → Fatigue

stage → events; the first mention fixes the stage; events at one stage co-exist

1.2 Hypothesis templates

Precedes	A tends to precede B
Co-occurs	A and B tend to co-occur
Sequence	A, then B, then C
First / Last	A is most often first (last)
Position	A falls in the earlier half
Proximity	A occurs right before B
Threshold	A precedes B in ≥ 2/3 of reports

1.3 Groups, families, levels

26 groups
template × scope: 9 report, 16 cohort, 1 two-cohort

7 claim families
precedes, co-occurs, sequence, first/last, position, proximity, threshold

3 difficulty levels
Easy: one report · Medium: one cohort · Hard: two cohorts

1.4 Gold annotations (computed mechanically)

Plan
- events, quantity
- denominator primitive, arms
- null kind, p0, tail
- test family, floor

Evidence sets
- denominator ids
- numerator ids
- excluded ids
- n, k per arm

Statistics, verdict
- p on both tails
- Wilson CI, Cohen's h
- supported / refuted / insufficient (sub-typed)

validated by six invariant families and a deterministic oracle; screened against permutation nulls

2 ART Verification Pipeline

from a natural-language claim to a verified verdict with auditable evidence

Inputs to the system

a claim in natural language, with its cohort and level

Auxiliary resources (to S1)

codebook: the corpus stipulations as text; null sources; test definitions; dataset card

annotation DB: report, vaccine, term, first-mention stage (ids pseudonymised)

S1 LLM — Articulate

form a structured plan; the claim-type mapping is withheld

1. **Parse the claim** — • events, cohort • quantity, direction
2. **Choose the primitive** — • denominator rule • number of arms
3. **Compose the plan** — • null kind, p0, tail • test family, floor
4. **Verify, 1 round** — • plan vs the codebook • re-ask once

Contract: the harness resolves the plan's primitive into its exact rule and states it to S2, so S2 implements a definition rather than interpreting advice

S2 LLM — Retrieve

plan to evidence sets

told the schema (column listing) and the contract; writes one SQL per arm, one row per report: in_den, in_num, excl

sandbox: read-only execution → n, k and report-id sets per arm

verify, up to 2 rounds: rejected SQL, n = k, empty arm, co-exist / tie / adjacency leaks

S3 function — Test

exact, deterministic

- exact binomial vs p0 (fixed, text-stated or data-derived)
- Fisher's exact test, two arms
- modal-vs-runner-up, argmax
- floor n ≥ 10; both tails

Supported **Refuted** **Insufficient**

insufficient: not-locatable / sparse / underpowered

3 Stage-level Evaluation

any stage can be set to gold to isolate the others

Plan correctness
the plan against the gold plan
plan EM, claim-type EM

Evidence correctness
the retrieved report sets against the gold sets
evidence F1, co-exist / tie leak, faithfulness P(evidence exact | verdict right)

Verdict and abstention
the test and verdict against the gold
accuracy, macro P/R/F1, p-agreement, overclaim, underclaim, sub-type accuracy